

# JeongBeom Baek (백정범)

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## Education

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| <b>Sungkyul University (성결대학교)</b> , Computer Engineering <ul style="list-style-type: none"> <li>1st Semester GPA: 3.97 / 4.5 (16 credits)</li> <li>C Programming: A+ (98)   AI Understanding &amp; Application: A+ (98)</li> <li>Basic Writing: A+ (95)   College Mathematics: B+ (87)</li> <li>Research &amp; Teaching Assistant under Prof. Hyeji Yang</li> <li>Nominated for university curriculum development initiative</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Anyang, South Korea<br>2026 – present |
| <b>Incheon Youngsun High School (인천영선고등학교)</b> , General Studies (Science Track) <ul style="list-style-type: none"> <li>Re-founded CodingLab (코딩랩) — delivered a deep learning theory curriculum using teacher training materials</li> <li>Information/CS: Rank 2 (consistently top performer across 3 years)</li> <li>Programming (elective): A grade — implemented NLP tokenization algorithms and Llama 3.2 Q&amp;A pipeline</li> <li>AI &amp; Physical Computing: Researched on-device AI chips, sLLM, and knowledge distillation to address LLM power consumption</li> <li>Mathematics &amp; Statistics: Derived backpropagation equations via chain rule/partial derivatives and analyzed AI output probability distributions</li> <li>Social Issues: Authored a report on AI hallucinations, dependence risks, and the necessity of AI literacy</li> <li>Data Analysis: Utilized Orange3 for data mining and GeoGebra for complex mathematical visualizations</li> <li>KSEF 2024: Published research on improving LLM reasoning via critical thinking frameworks</li> </ul> | Incheon, South Korea<br>2023 – 2025   |

## Experience

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| <b>Sungkyul University</b> , Undergraduate Student & Independent Researcher<br>Concurrently pursuing studies and participating in departmental initiatives. Independently archived and analyzed over 40 AI agent papers for an academic course assignment.                                                                                                                                                      | Anyang, South Korea<br>Mar 2026 – present<br>5 months            |
| <ul style="list-style-type: none"> <li>Exceeded academic assignment requirements by analyzing 40+ international papers on AI agents and RL</li> <li>Nominated to participate in a university curriculum development project</li> <li>Discussed Nunchi (눈치) humor-weight algorithm concept with Prof. Seong-jun Park (Dept. Head), resulting in an encouraging exploration of human-robot interaction</li> </ul> |                                                                  |
| <b>CodingLab (코딩랩)</b> , Lead Instructor & Re-founder<br>When the previous programming club dissolved, I stepped up to recreate the club as CodingLab. Delivered a deep learning theory curriculum using teacher training materials and MIT 6.S191. Conducted ~1 semester of lectures for students with no prior ML background.                                                                                 | Incheon Youngsun High School<br>Mar 2024 – Dec 2024<br>10 months |
| <ul style="list-style-type: none"> <li>Designed 7-week DL curriculum: perceptrons, backpropagation, CNNs, RNNs, Transformers, attention mechanisms</li> <li>Created original lecture materials (80+ slides) with mathematical rigor — equations, diagrams, code examples</li> <li>Mentored students from zero coding knowledge to understanding neural network architectures</li> </ul>                         |                                                                  |

## Incheon Youngsun High School, Coding Club Member & TA (코딩파티)

Active member of the school coding club. Helped peers with Python programming, developed apps using App Inventor, and served as a technical resource for classmates.

- Built custom shoe and eco-friendly fashion coordination apps using App Inventor, incorporating recommendation algorithms
- Coded a classroom seat arrangement program used by the entire class to solve real-world scheduling issues
- Tutored peers in Python (loops, functions, game programs) and served as a technical resource for classmates

Incheon, South Korea

Mar 2023 – Dec 2024

1 year 10 months

## Publications

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### Improving LLM Language Understanding through the Application of Critical Thinking

2024

Proposed applying human critical thinking patterns to LLM inference — generating and verifying logical hypotheses before producing responses, rather than direct token prediction. Fine-tuned Llama 3.2 3B with LoRA on a custom Korean dataset (merged-v2). The critical thinking model scored 0.566 vs baseline 0.466 on language understanding, reasoning, and requirement fulfillment metrics.

JeongBeom Baek

[ericbaek0828-netizen.github.io/assets/pdf/KSEF\\_2024\\_Paper.pdf](https://ericbaek0828-netizen.github.io/assets/pdf/KSEF_2024_Paper.pdf)

## Awards

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### 2026 Department Game Jam — High Commendation

July 2026

Built 'Escape from Uniformity' — a 2D side-scrolling adventure with suspicion gauge, stat-based NPC persuasion, and multi-ending structure. Used Harness Engineering (AI-assisted no-code workflow) as a freshman with a 3-person team over 4 days. Received highly positive verbal feedback for development execution.

Sungkyul University, Computer Engineering Dept.

## Skills

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### Programming & Engineering

### AI & Machine Learning

### Research & Teaching

### Tools & Platforms

## Languages

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### Korean

Native speaker

### English

Professional communication

## Interests

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### AI & Systems

### Knowledge Management

### Cognitive Science

## Projects

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### J.A.R.V.I.S.

2026 – present

Conceptual framework and architecture in active planning for a multi-agent orchestration system, focusing on layered agent routing, state management, and the Nunchi social context engine.

- Designing layered agent routing and automated quality gates for multi-agent workflows
- Nunchi algorithm: humor/sarcasm weight filtering based on unconscious physical signals
- Currently focusing on robust architectural planning and component design

### Escape from Uniformity (획일화 속 탈출)

June 2026 – July 2026

2026 Department Game Jam entry. A 2D side-scrolling adventure satirizing conformist society, with suspicion gauge, stat-based NPC persuasion, and multi-ending narrative structure.

- Harness Engineering workflow — AI-assisted no-code development as a freshman
- Perfect score in development category; judges called me ‘a developer to the bone’

### Critical Thinking LLM (KSEF 2024)

June 2024 – Nov 2024

Research project applying human critical thinking patterns to LLM inference. Fine-tuned Llama 3.2 3B to generate and verify hypotheses before responding.

- Custom Korean dataset (merged-v2) published on Hugging Face
- 21% improvement in language understanding score over baseline Llama 3.2